

Python Lecture 1

Efficient Exam-Ready Notes

1. Introduction to Python

- Python is a high-level programming language
- Used in: AI, Machine Learning, Data Science, Web Development
- Programming = writing instructions (code) for computers
- Computers understand binary (0s and 1s); Python translates to machine code
- Python code is executed using an interpreter

Advantages of Python

- Easy to learn, beginner-friendly syntax
- Open source and cross-platform
- Huge libraries ecosystem (pip, Django, etc.)

2. Python Setup

- Download from python.org
- Install VS Code (code editor)
- Verify installation:

```
python3 --version
```

First Program

```
print("Hello World")
```

Printing Multiple Items

```
print("Name:", "Shraddha")  
print("College:", "XYZ")
```

3. Variables

A variable stores a value in memory.

```
name = "Shraddha"  
age = 23  
price = 25.99
```

Variable Naming Rules

- Must start with a letter or underscore
- Cannot start with a number
- Cannot contain special characters (@ \$ % etc.)
- Can contain letters, digits, and underscores

Printing Variables

```
print("age")    # prints text: age
print(age)      # prints variable value: 23
```

4. Data Types

Data Type	Keyword	Example
Integer	int	23, -5
Float	float	3.14, 25.99
String	str	"Hello"
Boolean	bool	True, False
NoneType	None	None

Check type with:

```
print(type(age))    # <class "int">
```

5. Keywords

Reserved words with fixed meanings — cannot be used as variable names.

```
True   False   None   and   or   not   if   else   for   while
```

6. Case Sensitivity

Python is case-sensitive. Keywords must be written exactly.

```
True    (correct)      true    (ERROR)
age     Age     AGE     -- all different variables
```

7. Arithmetic Operators

Operator	Meaning	Example
+	Addition	a + b
-	Subtraction	a - b
*	Multiplication	a * b

Operator	Meaning	Example
/	Division (always float)	a / b
%	Modulus / Remainder	a % b
**	Power / Exponent	a ** b

Note: 4 / 2 = 2.0 -- Division ALWAYS returns float

8. Comments

Single Line

```
# This is a comment
```

Multi-line

```
"""
This is a multi-line comment
"""
```

Shortcut: Ctrl + / (Cmd + / on Mac)

9. Relational (Comparison) Operators

Compare two values and return True or False.

Operator	Meaning	Example
==	Equal	a == b
!=	Not equal	a != b
>	Greater than	a > b
<	Less than	a < b
>=	Greater or equal	a >= b
<=	Less or equal	a <= b

10. Assignment Operators

Operator	Meaning	Example
=	Assign	a = 5
+=	Add and assign	a += 5
-=	Subtract and assign	a -= 5
*=	Multiply and assign	a *= 5
/=	Divide and assign	a /= 5

Operator	Meaning	Example
<code>%=</code>	Modulo and assign	<code>a %= 5</code>
<code>**=</code>	Power and assign	<code>a **= 5</code>

11. Logical Operators

Operator	Meaning	Example
<code>not</code>	Reverses result (unary)	<code>not True => False</code>
<code>and</code>	Both must be True	<code>True and False => False</code>
<code>or</code>	At least one True	<code>True or False => True</code>

12. Type Conversion

Implicit (Automatic)

Python auto-converts lower to higher type in operations.

```
a = 5
b = 2.0
print(a + b)    # 7.0 (int promoted to float)
```

Explicit / Type Casting (Manual)

```
int()    float()    str()
a = int("5")        # 5
b = float("3.5")     # 3.5
int("hello")        # ERROR -- cannot convert letters
```

13. Input in Python

`input()` ALWAYS returns a **STRING** -- convert when needed.

```
name = input("Enter name: ")
age = int(input("Enter age: "))
marks = float(input("Enter marks: "))
```

14. Practice Programs

1. Sum of Two Numbers

```
a = int(input("Enter first number: "))
b = int(input("Enter second number: "))
print("Sum:", a + b)
```

2. Area of a Square

```
side = float(input("Enter side: "))
area = side * side
print("Area:", area)
```

3. Average of Two Numbers

```
a = float(input("Enter first number: "))
b = float(input("Enter second number: "))
avg = (a + b) / 2
print("Average:", avg)
```

4. Compare Two Numbers

```
a = int(input("Enter first number: "))
b = int(input("Enter second number: "))
print(a >= b)
```

Quick Revision Checklist

- What is Python
- Variables and naming rules
- Data types: int, float, str, bool, None
- Arithmetic, Relational, Logical, Assignment operators
- Input and output
- Type conversion (implicit and explicit)
- Comments